

Part A

1. a. ($\times$ 0 1)

b. (+ if (y, 0, y+x) set ($\times$ x p) 10)

c. Begin (while (x apply (x x)) x)

Part B

- 2 a. x is either a function parameter or a global variable
b. The value of the expression e is v

3 a. $\langle e, \xi, \phi, \rho \rangle \Downarrow \langle v, \xi, \phi, \rho \rangle$
b. $\langle e, \xi, \phi, \rho \rangle \Downarrow \langle v, \xi', \phi, \rho \rangle$

- 4 a. I think both are the same since after the while loop ended, the side effect for both is that the function parameter has changed and the value for the expression e is 0 for the global variable.
b. This is different because while the initial condition is the same, the conclusion is different since for the function parameter x is given value v in the original rule but only the environment for the function parameter changed.

$\text{set } (v \ 0)$	$\text{set } (v \ 0)$
$\text{set } (x \ v)$	$\text{set } (x \ 0)$
$(\text{define } y(x) (\text{set } x \ v))$	$(\text{define } y(x) (\text{set } v \ 1))$

Part C

a.
$$\frac{x \notin \rho \quad x \notin \xi \quad x \notin \phi \quad \langle e, \xi, \phi, \rho \rangle \Downarrow \langle v_1, \xi', \phi, \rho \rangle \quad v_1 = 0}{\langle \text{set}(x, v_1), \xi, \phi, \rho \rangle \Downarrow \langle v_1, \xi' \{x \mapsto v_1\}, \phi, \rho \rangle}$$

b.
$$\frac{x \notin \rho, x \notin \xi, x \notin \phi \quad \langle e, \xi, \phi, \rho \rangle \Downarrow \langle v_1, \xi, \phi, \rho' \rangle \quad v_1 = 0}{\langle \text{set}(x, v_1), \xi, \phi, \rho \rangle \Downarrow \langle v_1, \xi, \phi, \rho' \{x \mapsto v_1\} \rangle}$$

c. I prefer the second one since I can use the same var name in different contexts.

6a.

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(val x to
  (begin
    (define f (x y) x)
    (f 1 2)
    x))
```

b. Apply SimpleFixer fixes the bug by cleaning up the function parameters after the function returns restoring old variable values or deleting temporary ones so nothing "leaks" out of the function.